Mathematic Proof’s

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1 Introduction

A mathematical proof is a logical argument that establishes the truth of a mathematical statement. It uses established truths, definitions, and previously proven statements to logically demonstrate that a conclusion is valid. Proofs are crucial for verifying mathematical statements and building mathematical theories.

Terms:

- **Conjecture**: a statement that you think is true and can be proven (but hasn’t been proven yet).
- **Theorem**: a statement that has been shown to be true with a proof.
- **Proof**: a valid argument that shows that a theorem is true.
- **Premise**: a condition for the theorem, like ”if n is an even number...”.
- **Lemma**: a small theorem that we need to get the proof we’re interseted in.
- **Corollary**: a small theorem that follows from the more important one.

2 Disproving Conjectures

To disprove a conjecture, you need to find at least one counterexample, which is an example where the conjecture is false. A counterexample demonstrates that a conjecture is not always true.

3 Direct Proof

A direct proof is a way of showing the truth or falsehood of a given statement by a straightforward combination of established facts, usually axioms, existing lemmas and theorems, without making any further assumptions.

3.1 Example

If n is an even integer, then n^2 is also even.

Proof. Assume n is an even integer. By the definition of even numbers, there exists an integer k such that $n = 2k$.

Now square both sides:

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

Since $2k^2$ is an integer, n^2 is equal to 2 times an integer. Therefore, n^2 is even. □

4 Proof by Contraposition

Proof by Contraposition, is a rule of inference used in proofs, where one infers a conditional statement from its contrapositive.

4.1 Example

Proof. We will prove the contrapositive: *If n is odd, then n^2 is odd.*

Assume n is an odd integer. By the definition of odd numbers, there exists an integer k such that $n = 2k + 1$.

Now square both sides:

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2+2k$ is an integer, n^2 is equal to 2 times an integer plus 1. Therefore, n^2 is odd.

Thus, the contrapositive is true, and so the original statement holds: *If n^2 is even, then n is even.* □

5 Proof by Contradiction

Proof by Contradiction is based on a very simple principle: something that leads to a contradiction can not be true, and if so, the opposite must be true.

5.1 Example

Proof. We will prove the statement by contradiction.

Assume, for the sake of contradiction, that n^2 is even but n is odd.

Since n is odd, there exists an integer k such that $n = 2k + 1$.

Now square both sides:

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

This shows that n^2 is of the form $2m + 1$, which means n^2 is odd.

This contradicts our assumption that n^2 is even.

Therefore, our assumption is false, and it must be that if n^2 is even, then n is even. ☐

6 Proof by Cases

Proof by Cases is where a statement is proven by considering all possible cases or scenarios that could exist.

6.1 Example

Proof. We will prove the statement by considering two cases for the integer n: whether n is even or odd.

Case 1: n is even. Then there exists an integer k such that $n = 2k$. Squaring both sides:

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2),$$

which is even.

Case 2: n is odd. Then there exists an integer k such that $n = 2k + 1$. Squaring both sides:

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is odd.

So, if n is odd, then n^2 is odd. Therefore, if n^2 is even, n cannot be odd — it must be even.

Conclusion: If n^2 is even, then n is even. ☐

7 Existance Proof

An existance Proof is a demonstration that a particular mathematical object or property exists

7.1 Example

Proof. We want to prove that there exists an even integer n such that n^2 is divisible by 4.

Consider the even integer $n = 2$. Clearly, n is even.

Now compute:

$$n^2 = 2^2 = 4.$$

Since 4 is divisible by 4, this n satisfies the condition.

Therefore, there exists an even integer n such that n^2 is divisible by 4. ☐